

# Jason Ludmir

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## Education

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**Ph.D., Computer Science** *In-Progress*  
*Rice University*  
**Focus:** Quantum Computing and High Performance Computing, **Advisor:** Dr. Tirthak Patel

**M.Sc., Computer Science** *May 2021*  
*Brown University*  
**Focus:** Database Systems and Optimization, **Advisor:** Dr. Stanley Zdonik

**B.S., Finance** *May 2015*  
*Wake Forest University*  
**Focus:** International Business, **Minor:** Japanese Language

## Awards and Recognition

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**Rice University Nominee for the Google Ph.D. Fellowship** *2026*  
Selected as one of four graduate students across the university's graduate pool.

**Ken Kennedy – HPE Cray Graduate Fellowship** *2025*  
Selected for prestigious fellowship recognizing research excellence in high-performance computing.

**Best Lightning Talk at HPC Energy Conference** *2025*  
Awarded Best Lightning Talk by audience vote during the conference's lightning talk session.

**Nominee for the Nettie S. Autrey and Lodieska Stockbridge Vaughn Fellowships** *2025*  
The only student nominated in the Department of Computer Science at Rice University.

**1st Place: ACM Student Research Competition at SC 2024 Conference** *2024*  
Won after being selected to present at the prestigious ACM Student Research Competition.

**Inaugural HPC Student Cohort (HPCSC)** *2024*  
Selected at Supercomputing (SC) 2024 and IPDPS 2025 as part of a prestigious cohort promoting extended engagement with the HPC community.

**DAC Young Fellow** *2024*  
Selected as a Young Fellow of the 61st Design Automation Conference (DAC).

**DAC Creative Research Dissemination Award** *2024*  
Recognized for presenting a research publication summary with exceptional creativity.

**ICCAD Quantum Computing for Drug Discovery Challenge Honorable Mention and Innovation Award Nomination** *2023*  
Awarded for innovative contributions in applying quantum computing to a drug discovery challenge.

**Conference Travel Awards** *Multiple*  
Awarded travel awards to attend SC24, DAC24, SIGMETRICS25, ICS26.

**Ipreo International Hack-A-Thon: 1st Place** *2016*  
Led team to victory in a global competition focusing on innovative financial technology solutions.

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| <b>Bank of America Leadership Scholarship</b>                                     | 2014      |
| Recognized for outstanding leadership potential and academic excellence.          |           |
| <b>Global Citizens Scholarship</b>                                                | 2014      |
| Awarded for commitment to global awareness and cross-cultural understanding.      |           |
| <b>Wake Forest University Study Abroad Scholarship</b>                            | 2013      |
| Selected for academic merit to support international study experience in Japan.   |           |
| <b>Wake Forest University Dean's List</b>                                         | 2012-2015 |
| Consistently recognized for academic excellence throughout undergraduate studies. |           |

## Research Experience

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**QuEra Computing** February 2026 - May 2026  
*Scientific Software Intern*

- Developing algorithmic methods to improve utilization and execution efficiency on real digital neutral-atom quantum hardware.
- Building compiler- and scheduler-level optimizations for neutral-atom workloads across both logical and physical operation layers.
- Translating hardware constraints into practical software optimizations for neutral-atom systems in an industrial research setting.

**Rice University** August 2023 - Present  
*Doctoral Researcher*

- Advancing high-performance quantum computing through the development of optimized compilation techniques for Rydberg atom quantum computers, with a focus on hardware utilization.
- Developing innovative hybrid algorithms that combine machine learning and quantum computing techniques to tackle complex problems such as anomaly detection.

**Brown University** May 2020 - June 2021  
*Research Assistant*

- Researched optimizations to the dynamic time warping (DTW) algorithm in time series databases.
- Implemented optimizations to DTW in the Rust programming language.

## Peer-Reviewed Publications

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*In the computer science area, conference publications are considered one of the primary metrics for judging scholarly contributions. Conference publications have led to timely, impactful innovations due to the short turnaround time of the conference-based review system. Conference publications are rigorously peer-reviewed by experts in the field, and final decisions are made by a panel of experts. The acceptance rate of these conferences is typically less than 25%.*

1. **(ICS 2026, 21% acceptance rate)** Jason Ludmir, Nicholas S. DiBrita, Jason Han, and Tirthak Patel. "SpinTune: Improving the Reliability of Quantum Sensor Networks for Practical Quantum-Classical Utility." *In Proceedings of the ACM International Conference on Supercomputing (ICS)*, 2026.
2. **(ICS 2026, 21% acceptance rate)** Mohammad Abrarul Hasanat, Jason Ludmir, Tirthak Patel, and Rohan Basu Roy. "TuniQ: Autotuning Compilation Passes for Quantum Workloads

at Scale for Effectiveness and Efficiency.” *In Proceedings of the ACM International Conference on Supercomputing (ICS)*, 2026.

3. **(ICLR 2026, 28% acceptance rate)** Jason Han, Nicholas S. DiBrita, Daniel Leeds, Jianqiang Li, Jason Ludmir, and Tirthak Patel. “Layerwise Federated Learning for Heterogeneous Quantum Clients using Quorus.” *In Proceedings of the International Conference on Learning Representation (ICLR)*, 2026.
4. **(SIGMETRICS 2026, 21% acceptance rate)** Yuqian Huo, Daniel Leeds, Jason Ludmir, Nicholas S. DiBrita, and Tirthak Patel. “Anchor: Reducing Temporal and Spatial Output Performance Variability on Quantum Computers.” *In Proceedings of the ACM International Conference on Measurement and Modeling of Computer Systems (SIGMETRICS)*, 2026.
5. **(DAC 2025, 23% acceptance rate)** Jason Ludmir, Sophia Rebello, Jacob Ruiz, Tirthak Patel. “Quorum: Zero-Training Unsupervised Anomaly Detection using Quantum Autoencoders.” *In Proceedings of the Design Automation Conference (DAC)*, 2025.
6. **(SIGMETRICS 2025, 18% acceptance rate)** Jason Ludmir, Nicholas S. DiBrita, Yuqian Huo, Tirthak Patel. “Modeling and Simulating Rydberg Atom Quantum Computers for Hardware-Software Co-design with PachinQo.” *In Proceedings of the ACM International Conference on Measurement and Modeling of Computer Systems (SIGMETRICS)*, 2025.
7. **(SC 2024, 24% acceptance rate)** Jason Ludmir, Nicholas S. DiBrita, Yuqian Huo, Tirthak Patel. “PARALLAX: A Compiler for Neutral Atom Quantum Computers under Hardware Constraints.” *In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, 2024.
8. **(ICCAD 2024, 24% acceptance rate)** Nicholas S. DiBrita, Daniel Leeds, Yuqian Huo, Jason Ludmir, Tirthak Patel. “ReCon: Reconfiguring Analog Rydberg Atom Quantum Computers for Quantum Generative Adversarial Networks.” *In Proceedings of the International Conference on Computer-Aided Design (ICCAD)*, 2024.

### *Preprints*

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1. Jason Ludmir, Ian Martin, Nicholas S. DiBrita, and Tirthak Patel. “QuFoundry: Generating Data with Quantum Properties for Quantum Machine Learning Utility.”

### *Under-Review Manuscripts*

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1. **(Lead Author)** Park-n-Ride: A compilation and execution technique for Bivariate Bicycle (BB) codes on neutral atom architectures with scheduling and movements for functionalities such as shift automorphisms.
2. **(Lead Author)** PaQit: A compilation and architecture training for maximizing the number of parallel shots on neutral atom devices to increase energy efficiency and throughput while controlling fidelity loss.

### *Technical Reports*

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1. Jason Ludmir and Tirthak Patel. “CutSure: Reducing the Overhead of Reloading Rydberg Atom Arrays on Neutral Atom Quantum Computers.” ACM Report: <https://src.acm.org/binaries/content/assets/src/2025/jason-ludmir.pdf>

### *Open-Sourced Artifacts*

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1. **QuFoundry: Generating Data with Quantum Properties for Quantum Machine Learning Utility:** <https://github.com/positivetechnologylab/QuFoundry>.
2. **Layerwise Federated Learning for Heterogeneous Quantum Clients using Quorus:** <https://github.com/positivetechnologylab/Quorus>.
3. **Anchor: Reducing Temporal and Spatial Output Performance Variability on Quantum Computers:** <https://github.com/positivetechnologylab/Anchor>.
4. **Quorum: Zero-Training Unsupervised Anomaly Detection using Quantum Autoencoders:** <https://github.com/positivetechnologylab/Quorum>.
5. **CutSure: Increasing the Efficiency of Neutral Atoms by Reducing Qubit Waste from Measurement-related Ejections:** <https://github.com/positivetechnologylab/CutSure>.
6. **Modeling and Simulating Rydberg Atom Quantum Computers for Hardware-Software Co-design with PachinQo:** <https://github.com/positivetechnologylab/PachinQo>.
7. **PARALLAX: A Compiler for Neutral Atom Quantum Computers under Hardware Constraints:** <https://github.com/positivetechnologylab/Parallax>.  
<sup>†</sup>*Received Artifact Available, Functional, and Reproducible Badges at SC 2024.*
8. **ReCon: Reconfiguring Analog Rydberg Atom Quantum Computers for Quantum Generative Adversarial Networks:** <https://github.com/positivetechnologylab/ReCon>.

### *Speaking Events*

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| Presented “Quorum: Zero-Training Unsupervised Anomaly Detection using Quantum Autoencoders”,<br>DAC 2025                             | <i>June 2025</i>     |
| Presented “Modeling and Simulating Rydberg Atom Quantum Computers for Hardware-Software<br>Co-design with PachinQo”, SIGMETRICS 2025 | <i>June 2025</i>     |
| Poster Presentation, “Unsupervised Quantum Anomaly Detection”,<br>Quantum Information Processing Systems                             | <i>April 2025</i>    |
| Lightning Talk, “Unsupervised Quantum Anomaly Detection”,<br>Energy HPC Conference                                                   | <i>February 2025</i> |
| Poster Presentation, “Unsupervised Quantum Anomaly Detection”,<br>Energy HPC Conference                                              | <i>February 2025</i> |
| Poster Presentation, “Unsupervised Quantum Anomaly Detection”,<br>DefenseTech Connect Conference                                     | <i>December 2024</i> |
| Poster Presentation, “CutSure: Increasing the Efficiency of Neutral Atoms”,<br>ACM Student Research Competition at SC                | <i>November 2024</i> |
| Poster Presentation, “PARALLAX: Compiling for Neutral Atom Quantum Computers”,<br>Design Automation Conference (DAC)                 | <i>June 2024</i>     |
| Presented “Neutral Atom Compilation”, Rice University CS Graduate Seminar                                                            | <i>February 2024</i> |

### *Academic Service*

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| PC Member, <i>16th International Green and Sustainable Computing Conference</i> | <i>June 2026</i> |
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| Manuscript Reviewer, <i>Quantum Information Processing</i> (Quantum Computing)                                                                       | <i>July 2025</i>     |
| Manuscript Reviewer, <i>Cluster Computing</i> (Quantum Computing)                                                                                    | <i>June 2025</i>     |
| Manuscript Reviewer, <i>Quantum Machine Intelligence</i> (Quantum Computing)                                                                         | <i>May 2025</i>      |
| Judge, Shapiro Showcase<br>Shapiro Undergraduate Research Competition                                                                                | <i>April 2025</i>    |
| Poster Judge, Engineering and Computing Undergraduate Engineering and Computing<br>Engineering and Computing Undergraduate Research Poster Symposium | <i>February 2025</i> |
| Student Volunteer, Program Committee Meeting, High Performance Computing (HiPC)<br>Conference, HPC Systems Software Track                            | <i>August 2024</i>   |
| External Reviewer for quantum computing research, International Symposium<br>on Computer Architecture (ISCA)                                         | <i>January 2024</i>  |

### ***Mentorship Experience***

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| <b>Jacob Ruiz, Stanford University</b>                                                                                                                                                                                                                                                                                                                 | <i>July 2024 - November 2024</i>   |
| Guided implementation of quantum circuits for enhancing classical machine learning models with applications to anomaly detection in datasets. Co-author for Quorum (DAC 2025).                                                                                                                                                                         |                                    |
| <b>Sofia Rebello, Rice University</b>                                                                                                                                                                                                                                                                                                                  | <i>January 2024 - January 2025</i> |
| Mentored an undergraduate student in quantum computing applications in finance, demonstrating how quantum algorithms can optimize portfolio allocation and risk. Her poster on this work won first place at the ACM Student Research Competition (SRC) at the 2025 Code Generation and Optimization (CGO) conference. Co-author for Quorum (DAC 2025). |                                    |

### ***Teaching Experience***

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| <b>Rice University</b>                                                                                                                                                                                                                                                                            | <i>Fall 2024</i> |
| <i>Teaching Assistant - Database Management Systems</i>                                                                                                                                                                                                                                           |                  |
| <ul style="list-style-type: none"> <li>• Provided timely and thorough grading of assignments, delivering detailed, actionable feedback that reinforced students' understanding of core database concepts.</li> </ul>                                                                              |                  |
| <b>Rice University</b>                                                                                                                                                                                                                                                                            | <i>Fall 2023</i> |
| <i>Teaching Assistant - Distributed Programming</i>                                                                                                                                                                                                                                               |                  |
| <ul style="list-style-type: none"> <li>• Assisted in managing a highly collaborative class in which seniors prepared a capstone project.</li> </ul>                                                                                                                                               |                  |
| <b>Brown University</b>                                                                                                                                                                                                                                                                           | <i>Fall 2020</i> |
| <i>Head Teaching Assistant - Database Management Systems</i>                                                                                                                                                                                                                                      |                  |
| <ul style="list-style-type: none"> <li>• Led the remote redesign of one of the most popular courses at Brown, including creating assignments, exams, review materials and lectures.</li> <li>• Managed a team of TAs across the world in handling a student body of over 100 students.</li> </ul> |                  |
| <b>Brown University</b>                                                                                                                                                                                                                                                                           | <i>Fall 2019</i> |
| <i>Teaching Assistant - Computer Systems</i>                                                                                                                                                                                                                                                      |                  |
| <ul style="list-style-type: none"> <li>• Prepared assignments/labs using C and assembly for over 300 students enrolled.</li> <li>• Held weekly office hour sessions to assist students with debugging and conceptual questions.</li> </ul>                                                        |                  |

### ***Technical Skills***

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**Programming Languages:** Python, C, C++, Rust, Java, Assembly, LaTeX, SQL

**Frameworks & Libraries:** Matplotlib, NumPy, Pandas, Qiskit, SciPy

**Development Tools:** GDB, Git, Docker, VS Code, Jupyter Notebooks

**Parallel Computing:** CUDA, MPI, OpenMP

### *Graduate Coursework*

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#### **Rice University**

- Advanced Operating Systems
- Computer Systems Architecture
- Parallel Computing
- Probabilistic Algorithms
- Internet of Things (IoT)

#### **Brown University**

- Software Security and Exploitation
- Deep Learning
- Computer Systems
- Database Management Systems
- Time Series Databases

### *Additional Work Experience*

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#### **IHS Markit**

*May - August 2021*

*Data Science Intern*

- Optimized and debugged a machine learning model used to predict investor activity.

#### **Ipreo**

*May 2015 - August 2016*

*Market Intelligence Consultant*

- Provided clients with intelligence about institutional investors and drivers in equity markets.